DSA - PREFIX SUM

Agenda:

1. Prefix Sum
2. Sum of Even/odd Indexed Element in a given range.
3. Special Index Problem
4. Count of even numbers in a given range

---

1 Prefix Sum O(N) T.C

```
int n = A.length
int[] prefix = new int [n]; // array to store prefix sums
prefix[0] = A[0]    //(Base Case) (first prefix value is same as 1st element
    //Build prefix sum for every index i.
for (int i = 1 ; i<n ; i++){
    prefix [i] = prefix[i-1] + A[i]    // Prefix[i] = sum of all elements from A[0] to A[i]
                                       // we add curr. element A[i] to the previous prefix sum.
}
return prefix;
}
```

---

2 Sum of Odd Indexed element in range (By GPT (no-error) approach)

- - - - (int[] A, int[][] B)

```
int oSum [] = new int [A.length +1];
for ( int i=0 ; i< A.length ; i++){
    if (i% 2 != 0) {
        oSum[i+1] = oSum [i] + A[i];
    else {
        oSum [i+1] = oSum [i]
        }
    }
int s=0;
int e=0;
int sum[] = new int [B.length];
for (int i=0 ; i < B.length; i++ {
        s= B[i][0] ;   e = B [i][1];
        sum [i] = oSum [e+1] - oSum[s] ;
}
```

③ Special Index Problem

Count how many indices i such that removing A[i] results in sum of even-indexed elements = sum of odd-indexed elements

```
int n = A.length
int [] PE = new int [n+1];
int [] PO = new int [n+1];
for (int i=0 ; i<n; i++) {
        if ((i % 2) == 0) {
                PE[i+1] = PE[i] + A[i]
                PO[i+1] = PO[i];
        }
        else {
                PO[i+1] = PO[i] + A[i];
                PE[i+1] = PE[i];
        }
}
```

(even index) add to PE

(odd index) add to PO

Final:

PE = [0, 2, 2, 8, 8]

PO = [0, 0, 1, 1, 5]

```
int count = 0;
for (int i=0; i<n; i++) {
    int S_even = PE[i] + (PO[n] - PO[i+1]);
    int S_odd = PO[i] + (PE[n] - PE[i+1]);
    if (S_even = S_odd) {
        count++;
    }
}
```

④ Count of even numbers in range

same like ② just, replace A[i] with 1.

```
eCount[i+1] = eCount[i] + 1;
```